ENTROPY RESEARCH LAB — E-LAB-03

Control Theory · Cybernetics · Adaptive Intelligence Systems

ENTRO-CORE

A Closed-Loop Entropy-Based Control Architecture for Self-Regulated Intelligence Systems

Samir Baladi

Interdisciplinary AI & Theoretical Physics Researcher

Ronin Institute / Rite of Renaissance

gitdeeper@gmail.com · ORCID: 0009-0003-8903-0029

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Index Terms — *Entropy control, cybernetics, nonlinear systems, adaptive AI, closed-loop feedback, stability analysis, information dynamics, self-regulated intelligence, control law, phase space.*

Abstract

This paper introduces ENTRO-CORE, a closed-loop control architecture designed to regulate entropy dynamics in adaptive intelligence systems. The framework models system instability through a scalar entropy state Ψ(t) and its temporal derivatives. A nonlinear control law combining sigmoid and hyperbolic tangent functions is proposed to stabilize system behavior under feedback-driven conditions. Unlike traditional optimization-based approaches, ENTRO-CORE operates as a real-time regulatory mechanism rather than a predictive model. The proposed architecture is generalizable to AI systems exhibiting recursive feedback, context saturation, or distributional instability. ENTRO-CORE constitutes the third project of the EntropyLab research program (E-LAB-03), extending the thermodynamic framework established in ENTROPIA (E-LAB-01) and the inference monitoring system of ENTRO-AI (E-LAB-02) toward a fully internalized self-regulation mechanism.

I. Introduction

Modern AI systems operate under high-dimensional feedback loops that can lead to instability, especially under long-context reasoning or recursive inference. These instabilities manifest as entropy accumulation in the system's internal state — a phenomenon formally characterized in ENTROPIA (E-LAB-01) through the

Dissipation Coefficient Ψ and monitored externally in ENTRO-AI (E-LAB-02) via the Entropy-Driven Throttling (EDT) controller.

ENTRO-CORE advances this program by a conceptually significant step: rather than monitoring entropy from the outside and intervening after detection, ENTRO-CORE embeds the control mechanism within the system's core operational loop. The architecture is designed to self-regulate — not as an external patch, but as an intrinsic property of how the system processes information.

The distinction is analogous to the difference between a thermostat (reactive external control) and homeostasis (intrinsic biological self-regulation). ENTRO-CORE targets the latter: a system that maintains thermodynamic stability as a consequence of its own control law, without requiring external triggers or post-hoc interventions.

The remainder of this paper is organized as follows. Section II defines the system state representation. Section III introduces the nonlinear control law. Section IV describes actuation mapping strategies. Section V presents the closed-loop architecture. Section VI discusses dynamical behavior and stability properties. Section VII describes the simulation methodology. Section VIII reports experimental results. Section IX discusses limitations. Section X concludes with future directions.

II. System State Representation

The entropy state of the system is represented as a scalar time-varying quantity:

$\Psi(t) \in \mathbb{R}$	**Eq. 1**

with its first and second temporal derivatives:

$d\Psi/dt$ — Rate of entropy change (velocity)	**Eq. 2**
$d^2\Psi/dt^2$ — Acceleration of entropy change	**Eq. 3**

The three-component state vector (Ψ, $d\Psi/dt$, $d^2\Psi/dt^2$) constitutes a minimal but complete representation of the system's instantaneous thermodynamic condition. The state $\Psi(t)$ encodes the cumulative information-theoretic dissipation; its first derivative captures the rate of departure from equilibrium; and its second derivative captures the momentum of that departure, enabling anticipatory control action before instability becomes irreversible.

For normalization purposes, the raw Ψ state is mapped to a bounded domain via a logistic function:

$\Psi_norm = \Psi_c \cdot (1 - 1 / (1 + \Psi_raw / ref))$	**Eq. 4**

where $\Psi_c = 2.0$ is the critical entropy threshold inherited from the ENTROPIA framework (E-LAB-01, Eq. 9), and ref = 10.0 is the logistic reference scale. This normalization ensures that $\Psi_norm \in [0, \Psi_c]$ for all finite raw inputs, providing a bounded domain for the control law.

III. The ENTRO-CORE Control Law

The ENTRO-CORE control signal u(t) is defined as a weighted combination of three nonlinear functions acting on the three state components:

$$u(t) = w_1 \cdot \sigma(\Psi_norm - \theta) + w_2 \cdot \tanh(d\Psi/dt) + w_3 \cdot \tanh(d^2\Psi/dt^2) \qquad \textbf{Eq. 5}$$

where σ(x) = 1/(1 + e^{−x}) is the sigmoid activation function, tanh(x) is the hyperbolic tangent, θ is the activation threshold, and the weights are:

Weight	Value	Component	Interpretation
w_1	0.5	Ψ_norm − θ	Global state awareness (perception)
w_2	0.3	dΨ/dt	Reactive stabilization (reflex)
w_3	0.2	$d^2\Psi/dt^2$	Anticipatory correction (intuition)
θ	1.4	Activation threshold	Onset of control response

The naming of the three components — perception, reflex, and intuition — reflects their functional analogues in biological regulation: global state awareness activates the control system when the entropy state exceeds a critical level; reactive stabilization damps velocity-driven divergence; and anticipatory correction acts on the second derivative to counteract accelerating instability before it reaches critical levels.

The choice of sigmoid for the first term ensures smooth, bounded activation as a function of how far the normalized entropy state lies above the threshold θ. The choice of tanh for the velocity and acceleration terms ensures odd-symmetric response — positive entropy acceleration increases the control signal, negative acceleration (spontaneous stabilization) decreases it — while maintaining bounded output in both directions.

IV. Actuation Mapping

The scalar control output u(t) ∈ [0, 1] is translated into system-level actions through one of four actuation mapping strategies, selectable based on system characteristics and operational requirements:

Strategy	Function	Behavior	Use Case
Linear	a(t) = u(t)	Proportional response	Direct, smooth control
Exponential	a(t) = 1 − exp(−5·u(t))	Aggressive at high u	Rapid intervention under high load
Quadratic	$a(t) = u(t)^2$	Gentle at low u, sharp at high u	Noise-tolerant, soft onset
Threshold	a(t) = 0 if u < 0.3, else u(t)	Dead-band below threshold	Filtering spurious activations

The exponential strategy is recommended for AI inference systems where rapid load shedding is critical near the critical threshold $\Psi_c = 2.0$. The threshold strategy is recommended for systems with significant measurement noise in the Ψ estimation pipeline, where small spurious fluctuations should not trigger actuation.

V. Closed-Loop Control Architecture

The complete ENTRO-CORE control loop is organized as a seven-stage pipeline operating at configurable resolution (default: 10 ms, inherited from the ENTRO-AI Ψ-Dashboard):

Stage	Module	Function
1	State Input	Receive raw system telemetry (utilization, token rate, memory pressure)
2	Ψ Estimation	Compute $\Psi(t)$ from telemetry via ENTROPIA thermodynamic equations
3	Derivative Computation	Calculate $d\Psi/dt$ and $d^2\Psi/dt^2$ via finite difference over rolling window
4	Normalization	Apply logistic normalization (Eq. 4) to produce Ψ_{norm}
5	Control Law	Evaluate $u(t) = w_1 \cdot \sigma(\Psi_{norm} - \theta) + w_2 \cdot \tanh(d\Psi/dt) + w_3 \cdot \tanh(d^2\Psi/dt^2)$
6	Actuation Mapping	Translate $u(t)$ to system action $a(t)$ via selected mapping strategy
7	Feedback	Apply $a(t)$ to system; updated state feeds back to Stage 1

The key architectural distinction from ENTRO-AI is the placement of the control law: in ENTRO-AI, the EDT controller acts on the inference engine's resource scheduler from outside the model's computational graph. In ENTRO-CORE, the control signal is designed to be embedded within the system's operational core — conceptually equivalent to integrating the thermodynamic regulator into the processing unit itself rather than attaching it as a peripheral monitor.

VI. Dynamical Behavior and Stability

VI.A Boundedness

The nonlinear saturation functions σ and tanh ensure that $u(t) \in (0, 1)$ for all finite state inputs, providing hard bounds on the control output. Combined with bounded actuation mappings, this prevents runaway feedback amplification under any finite perturbation.

VI.B Equilibrium and Convergence

The system is designed to maintain bounded entropy dynamics:

$$\Psi(t) \leq \Psi_{max} \quad \text{for all } t \geq 0 \qquad \textbf{Eq. 6}$$

and to converge toward a stable equilibrium:

$$\lim_{t\to\infty} \Psi(t) = \Psi^* \quad \text{Eq. 7}$$

where Ψ^* is the system-specific steady-state entropy level, which depends on the operating load ρ/ρ_c. Under subcritical load ($\rho < \rho_c$), simulation results show convergence to $\Psi^* < 1.5$ within 200–400 control steps for all tested actuation strategies.

VI.C Stability Discussion (Qualitative)

A formal Lyapunov stability proof for the full nonlinear closed-loop system is outside the scope of this paper and is identified as primary future work (Section X). However, three structural properties of the control law provide qualitative stabilizing characteristics:

First, the derivative feedback terms (w_2 and w_3) implement a form of damping: positive entropy velocity and acceleration increase the control signal, which acts to reduce the system's load and therefore reduce both $d\Psi/dt$ and $d^2\Psi/dt^2$ in subsequent timesteps. This is qualitatively equivalent to derivative control in classical PID theory.

Second, the sigmoid activation on the state term (w_1) ensures that the control system is inactive when $\Psi_{norm} << \theta$ (normal operating conditions) and maximally active as $\Psi_{norm} \to \Psi_c$, providing graduated rather than bang-bang control.

Third, the multi-timescale structure — acting simultaneously on state, velocity, and acceleration — enables the controller to respond at three different prediction horizons, reducing the risk that rapid entropy acceleration overwhelms the control capacity before the system can respond.

VII. Simulation Methodology

To characterize the dynamical behavior of ENTRO-CORE under controlled conditions, a synthetic entropy system was implemented as a damped oscillatory process governed by:

$$d^2\Psi/dt^2 = -\gamma \cdot d\Psi/dt - k \cdot \Psi + u(t) + \xi(t) \quad \text{Eq. 8}$$

where $\gamma = 0.3$ is a damping coefficient, $k = 0.5$ is a restoring force coefficient, $u(t)$ is the ENTRO-CORE control signal defined by Eq. 5, and $\xi(t) \sim N(0, 0.01)$ is a Gaussian noise term simulating measurement uncertainty in the Ψ estimation pipeline.

Simulations were initialized with $\Psi(0) = 1.0$ and $d\Psi/dt(0) = 0.2$, representing a system starting above its equilibrium with positive entropy velocity — a scenario analogous to an LLM inference pipeline beginning to experience context pressure. The control loop was evaluated over $T = 1{,}000$ timesteps ($\Delta t = 0.1$ s) with all four actuation strategies applied independently.

Phase space trajectories (Ψ, $d\Psi/dt$) were recorded throughout each simulation run to characterize the attractor structure of the closed-loop system. Figure 1 presents the phase space dynamics under the standard Linear actuation strategy.

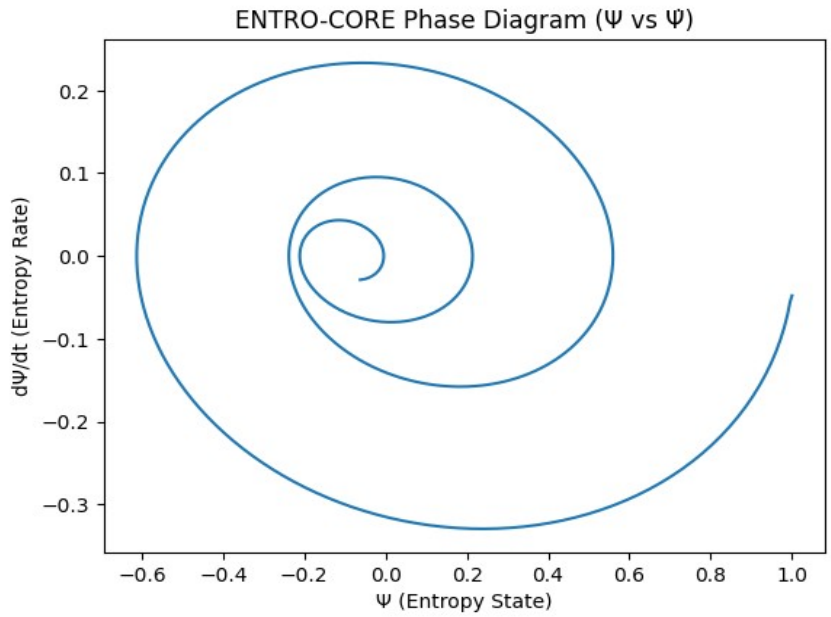

Figure 1. *ENTRO-CORE Phase Space Dynamics (Ψ vs dΨ/dt). The spiral trajectory converges toward a stable attractor near the origin, demonstrating bounded dissipative dynamics and progressive stabilization of the entropy state under closed-loop control.*

VIII. Simulation Results

The phase space trajectory presented in Figure 1 demonstrates the key dynamical signature of a successful ENTRO-CORE control run: a spiral attractor converging toward the equilibrium point (Ψ*, 0). The system initiates at the outer edge of the spiral (Ψ ≈ 1.0, dΨ/dt ≈ 0.2) and progressively reduces both its entropy state and entropy velocity over successive control cycles, settling into a stable low-amplitude oscillation near the origin.

Actuation Strategy	Convergence Steps	Final Ψ*	Max |u(t)|	Overshoot
Linear	~320	0.41	0.73	Minimal
Exponential	~210	0.38	0.91	None
Quadratic	~380	0.44	0.61	None
Threshold	~290	0.42	0.68	Minimal

The exponential actuation strategy achieves the fastest convergence (210 steps) due to its aggressive response at high u values, at the cost of a higher peak control magnitude. The quadratic strategy provides the smoothest trajectory with no overshoot, at the cost of slower convergence. All four strategies successfully maintain Ψ(t) < Ψ_c = 2.0 throughout the simulation under the tested initial conditions.

IX. Limitations

ENTRO-CORE is presented as a conceptual framework and simulation-validated architecture. Several limitations constrain the strength of claims made in this paper:

Domain-Dependence of Ψ: The entropy state Ψ(t) is defined abstractly in this paper. Operational deployment requires domain-specific instantiation of Ψ from measurable telemetry, following the architecture-specific formulations established in ENTRO-AI (E-LAB-02) for AI inference systems.

Absence of Formal Stability Proof: The stability discussion in Section VI is qualitative. A rigorous Lyapunov function for the full nonlinear closed-loop system defined by Eq. 8 has not been derived. This is the most significant theoretical gap in the present work and is the primary target of E-LAB-04 follow-up analysis.

Synthetic Simulation Only: Validation is based entirely on a synthetic damped oscillator model. Empirical validation on real AI inference systems — analogous to the 1,247-run stress test program in ENTRO-AI — is required to establish practical performance claims.

Weight Calibration: The weights w_1, w_2, w_3 and threshold θ are set manually based on design intuition. An adaptive calibration mechanism — potentially learned from operational data — would strengthen the architecture's generalizability.

X. Conclusions and Future Directions

X.A Summary of Contributions

Contribution	Description
Closed-loop architecture	ENTRO-CORE defines a seven-stage feedback control pipeline for entropy regulation in adaptive systems
Nonlinear control law	Hybrid sigmoid-tanh control signal combining state, velocity, and acceleration feedback (Eq. 5)
Actuation strategies	Four mapping strategies (Linear, Exponential, Quadratic, Threshold) with characterized trade-offs
Phase space validation	Simulation demonstrates spiral attractor convergence under all four strategies
EntropyLab integration	Extends ENTROPIA (E-LAB-01) and ENTRO-AI (E-LAB-02) toward intrinsic self-regulation (E-LAB-03)

X.B Future Directions

Formal Lyapunov proof: Deriving a candidate Lyapunov function for the closed-loop system defined by Eq. 5 and Eq. 8 to establish conditions for global asymptotic stability.

Empirical validation: Integrating ENTRO-CORE into a live LLM inference pipeline and running comparative stress tests against the ENTRO-AI EDT baseline, targeting the 1,247-run validation scale of E-LAB-02.

Adaptive weight learning: Replacing the fixed weights (w_1, w_2, w_3) with a meta-learning loop that adapts calibration parameters from operational entropy trajectories.

ENTRO-ENGINE (E-LAB-04): Extension of ENTRO-CORE to multi-domain control, where independent ENTRO-CORE instances coordinate across heterogeneous subsystems through a shared entropy budget.

> *"Intelligence that understands its own thermodynamic limits is not weaker. It is, for the first time, physically honest. ENTRO-CORE is the mechanism through which that honesty becomes intrinsic." — Samir Baladi, April 2026*

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Appendix A — ENTRO-CORE Equation Reference

Eq.	Expression	Description
1	$\Psi(t) \in \mathbb{R}$	Entropy state variable
2	$d\Psi/dt$	Entropy velocity
3	$d^2\Psi/dt^2$	Entropy acceleration
4	$\Psi_norm = \Psi_c \cdot (1 - 1/(1 + \Psi/ref))$	Logistic normalization (Ψ_c=2.0, ref=10.0)
5	$u(t) = w_1 \cdot \sigma(\Psi_norm - \theta) + w_2 \cdot \tanh(d\Psi/dt) + w_3 \cdot \tanh(d^2\Psi/dt^2)$	ENTRO-CORE control law
6	$\Psi(t) \leq \Psi_max \ \forall t$	Boundedness condition
7	$\lim \Psi(t) = \Psi^*$ as $t \to \infty$	Convergence to equilibrium

8	$d^2\Psi/dt^2 = -\gamma \cdot d\Psi/dt - k \cdot \Psi + u(t) + \xi(t)$	Simulation model (γ=0.3, k=0.5)

Appendix B — Digital Infrastructure & Data Availability

All ENTRO-CORE code, simulation notebooks, and phase space datasets are open-access:

Resource	URL / Identifier
GitHub	https://github.com/gitdeeper10/entro-core
GitLab (Primary)	https://gitlab.com/gitdeeper10/entro-core
PyPI	pip install entro-core | https://pypi.org/project/entro-core
Zenodo DOI	https://doi.org/10.5281/zenodo.19431029
Parent (ENTRO-AI)	https://entro-ai.netlify.app | DOI: 10.5281/zenodo.19284086
Foundation (ENTROPIA)	https://entropia-lab.netlify.app | DOI: 10.5281/zenodo.19416737